ALLISON LAU

+1 (416) 817-8650
allison.lau@mail.utoronto.ca

Education

2021 - **University of Toronto**, Faculty of Arts and Science

Honours Bachelor of Science, GPA 3.94/4.0
Double Specialists: Physics, Computer Science (Focus in Artificial Intelligence)
Minor: Mathematics
Courses: Electricity and Magnetism, Intro. to Quantum Mechanics, Thermal Physics, Software Design, Computer Organization, Software Tools and Systems Programming, Data Structure and Analysis

Work/Experience

May 2023 - **University of Toronto Dunlap Institute Optical Instrumentation Lab** | Research Assistant

- Developed and implemented object-oriented programming (OOP) in Python to automate the testing of CMOS detectors used for space imaging
- Developed an opto-mechanical alignment and testing system for fiber characterization

Apr 2023 **Breakout Game Project**

- Created a breakout game using assembly language
- Demonstrated proficiency in low-level programming and computer architecture

Feb 2023 **Car Battery Prediction Project**

- Implemented data cleaning, employed clustering algorithms, random forests, and linear regression models to identify patterns
- Ranked 7th in the University of Toronto D3S Datathon

Nov 2022 **Texas Hold'em Game Project**

- Worked with a team of 8 members to develop a Java-based game
- Used clean architecture and design patterns to create a scalable and efficient application
- Demonstrated software engineering skills by implementing features such as game logic, user interface, and data persistence

Skills

Programming	Python [Pandas / NumPy / Matplotlib / SciPy / SymPy / scikit-learn] | Java [Swing / IO] | C / C++ | LaTeX | Kotlin | Assembly | sh | MATLAB
Web dev.	HTML / CSS / JavaScript
Data vis.	R | Figma | Tableau
Others	3D Printing, Digital Electronics, Machining

Groups

2022-2023 Vice President, University of Toronto Hong Kong Public Affairs and Social Services Society

- Chief Editor for the annual publication
- Collaborated with community partners to develop and execute initiatives

2022-2023 Vice President, University of Toronto Cantonese Debate Society

- Led meetings and organized events for a team of 10 members
- Coached members on debate skills and facilitated team building activities